

HSS model of a Buck-boost converter

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Contents

1	Buck-boost converter theoretical study	2
1.1	Buck-boost topology	2
1.2	Buck-boost operation sequences	3
2	Modeling methods	5
2.1	Time State-Space detailed model (TSS)	5
2.2	Time State-Space average model	6
2.3	Harmonic State-Space detailed model (HSS)	8
2.3.1	Matrix X	9
2.3.2	Matrix U	9
2.3.3	Matrix $\mathcal{A}$	10
2.3.4	Matrix $\mathcal{B}$	12
2.4	Harmonic State-Space average model	13
2.4.1	Matrix X_{ave}	13
2.4.2	Matrix U_{ave}	14
2.4.3	Matrix $\mathcal{A}_{ave}$	14
2.4.4	Matrix $\mathcal{B}_{ave}$	15
3	Simulation and results	17
3.1	Detailed models comparison	17
3.1.1	Parameters choice	18
3.1.2	Simulation results	19
3.2	Average models comparison	20
3.2.1	Simulation results	21

Chapter 1

Buck-boost converter theoretical study

In this chapter, the Buck-boost converter operation is described in order to deduce the equations that represent its functioning, mainly during the steady-state phase.

1.1 Buck-boost topology

The Buck-boost topology is shown in [Figure 1.1](#). It is composed of:

- Two semiconductor switches that dictates the converter operation, K1 is a controllable switch such as a MOSFET and K2 is a diode.
- Inductor L used for energy storage used when K1 is off.
- Capacitor C used for maintaining the voltage on the load with its energy storage when K1 is on and filtering switching effects.

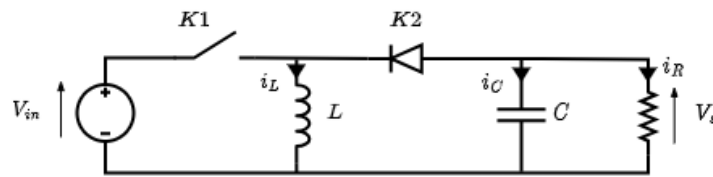


Figure 1.1: Buck-boost circuit topology.

For this study, we will consider the following two hypotheses such as done for the Buck HSS model study [\[1\]](#):

- The converter operates in continuous conduction mode, which means that the switching frequency is high enough to avoid de-charging the inductor L completely.
- The converter switches $K1$ and $K2$ are ideal.

1.2 Buck-boost operation sequences

The Buck-boost converter has two distinct operation sequences under the continuous conduction mode shown in Figure 1.2 and Figure 1.3, where α is the duty cycle and $T_{sw} = \frac{1}{f_{sw}}$ is the switching period.

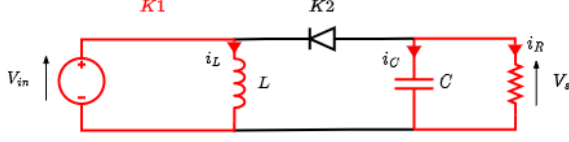


Figure 1.2: Sequence 1: K1 on, K2 off

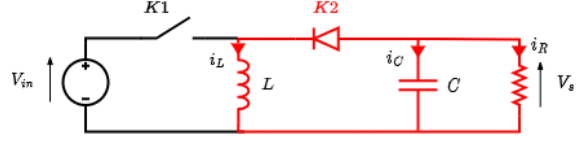


Figure 1.3: Sequence 2: K1 off, K2 on

Sequence 1: $0 < t < \alpha T_{sw}$

Figure 1.2 circuit shows how the converter operates during this sequence, where K1 (MOSFET) is on and K2 (diode) is off.

During this sequence, the inductor is magnetized by the DC voltage source, therefore:

$$L \frac{d}{dt} i_L = V_{in} \quad (1.1)$$

Integrating (1.1) gives us the expression of i_L during sequence 1:

$$i_L(t) = \frac{V_{in}}{L} t + cst \quad (1.2)$$

In the continuous conduction mode, we consider that i_L doesn't reach 0 and has a minimum value I_{min} . Assuming that i_L has its minimum value at $t = 0$, $i_L(0) = I_{min}$ and i_L expression becomes:

$$i_L(t) = \frac{V_{in}}{L} t + I_{min} \quad (1.3)$$

Due to the inductance magnetization by the DC power supply, at $t = \alpha T_{sw}$ the current i_L will reach its maximum value I_{max} . Therefore:

$$I_{max} = \frac{V_{in}}{L} \alpha T_{sw} + I_{min} \quad (1.4)$$

By definition, the current amplitude is then equal to:

$$\Delta I = I_{max} - I_{min} = \frac{V_{in}}{L} \alpha T_{sw} \quad (1.5)$$

Finally, during sequence 1, the voltage output is maintained by the capacitor, such that:

$$C \frac{d}{dt} V_s = i_c \quad (1.6)$$

Since $i_c = -i_R$ and $i_R = \frac{V_s}{R}$:

$$C \frac{d}{dt} V_s = -\frac{V_s}{R} \quad (1.7)$$

Sequence 2: $\alpha T_{sw} < t < T_{sw}$

Figure 1.3 circuit shows how the converter operates during this sequence, where K1 (MOSFET) is off and K2 (diode) is on.

During this sequence, the inductor is demagnetized while providing current to the load. From the circuit of Figure 1.3:

$$L \frac{d}{dt} i_L = V_s \quad (1.8)$$

Integrating (1.8) gives us the expression of i_L during sequence 2:

$$i_L(t) = \frac{V_s}{L}(t - \alpha T_{sw}) + cst \quad (1.9)$$

In the continuous conduction mode, assuming that at $t = \alpha T_{sw}$ the current i_L reach its maximum value I_{max} , $i_L(\alpha T_{sw}) = I_{max}$ and i_L expression becomes:

$$i_L(t) = \frac{V_s}{L}(t - \alpha T_{sw}) + I_{max} \quad (1.10)$$

Due to the inductance demagnetization, at $t = T_{sw}$ the current i_L will reach its minimum value I_{min} again. Therefore:

$$I_{min} = \frac{V_s}{L}(1 - \alpha)T_{sw} + I_{max} \quad (1.11)$$

By definition, the current amplitude is equal to:

$$\Delta I = I_{max} - I_{min} = -\frac{V_s}{L}(1 - \alpha)T_{sw} \quad (1.12)$$

By equalizing the two current amplitude equations found (1.5) = (1.12), we find the relation between the input and output voltages:

$$V_s = -\frac{\alpha}{(1 - \alpha)}V_{in} \quad (1.13)$$

Therefore, $-\frac{\alpha}{(1 - \alpha)}$ is the Buck-boost converter gain. Notice that the output voltage polarity is reversed and that the converter can act as a Buck or a Boost depending on the duty cycle value:

$$\begin{cases} \alpha < 0.5, & \text{Buck mode} \\ \alpha > 0.5, & \text{Boost mode} \end{cases} \quad (1.14)$$

Chapter 2

Modeling methods

In this chapter, the steps to achieve the HSS model of the Buck-boost converter are described. For this, we take for beginning the equations of the dynamics of the Buck-boost converter developed in [chapter 1](#).

2.1 Time State-Space detailed model (TSS)

To implement a State-Space representation of the Buck-boost converter that considers the switching of K1 and K2, the switching function $s(t)$ is defined as:

$$s(t) = \begin{cases} 1, & \text{then } K_1 \text{ is ON and } K_2 \text{ is OFF,} \\ 0, & \text{then } K_1 \text{ is OFF and } K_2 \text{ is ON.} \end{cases} \quad (2.1)$$

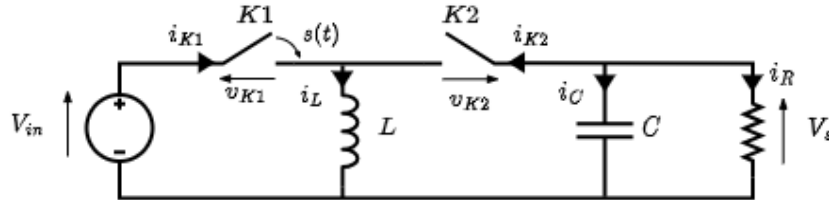


Figure 2.1: Buck-boost circuit with respect to switching function.

Using the conventions defined in [Figure 2.1](#), the switches voltages and currents expressions in function of $s(t)$ are:

$$\begin{aligned} i_{k1} &= s(t) \cdot i_L \\ i_{k2} &= (1 - s(t)) \cdot i_L \\ v_{k1} &= (1 - s(t)) \cdot (V_{in} - V_s) \\ v_{k2} &= -s(t) \cdot (V_{in} - V_s) \end{aligned} \quad (2.2)$$

By manipulating these equations, we obtain that

$$\begin{aligned} i_{k1} + i_{k2} &= i_L \\ v_{k1} - v_{k2} &= V_{in} - V_s \end{aligned} \quad (2.3)$$

By applying the KCL and the KVL, we obtain the inductor voltage and the capacitor current expressions:

$$\begin{aligned} i_c &= -i_{k_2} - i_R \\ v_L &= -v_{k_2} + V_s = -v_{k_1} + V_{in} \end{aligned} \quad (2.4)$$

Therefore, by applying (2.4) on the capacitor and inductor equations, we obtain the following equation system that represents the detailed converter operation:

$$\begin{cases} C \frac{d}{dt} V_s = i_C = -(1 - s(t))i_L - i_R \\ L \frac{d}{dt} i_L = v_L = s(t) \cdot (V_{in} - V_s) + V_s \end{cases} \quad (2.5)$$

Using that $i_R = \frac{V_s}{R}$ and manipulating the first and second equations:

$$\begin{cases} \frac{d}{dt} V_s = \frac{s(t)-1}{C} i_L - \frac{1}{RC} V_s \\ \frac{d}{dt} i_L = \frac{1}{L} s(t) \cdot V_{in} + \frac{1-s(t)}{L} V_s \end{cases} \quad (2.6)$$

From the system (2.6), the state-space representation can be written as follows:

$$\frac{d}{dt} x(t) = A(t)x(t) + B(t)u(t) \quad (2.7)$$

with,

$$x = \begin{bmatrix} i_L \\ V_s \end{bmatrix} \quad (2.8)$$

$$u = s(t) \cdot V_{in} \quad (2.9)$$

$$A = \begin{bmatrix} 0 & \frac{1-s(t)}{L} \\ \frac{s(t)-1}{C} & -\frac{1}{RC} \end{bmatrix} \quad (2.10)$$

$$B = \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} \quad (2.11)$$

Notice that this state-space representation considers the switching effect. We can see that the matrix A and the input u are dependent of the switching function, which will be extremely important during the implementation of the HSS model.

2.2 Time State-Space average model

The average state-space representation of the Buck-boost converter consists of taking the average of the precedent state-space system over one switching period.

The average value of a periodic signal is defined as:

$$\langle x \rangle_T(t) = \frac{1}{T} \int_t^{t+T} x(\tau) d\tau \quad (2.12)$$

By performing this operation, we can neglect the part of the periodic signals related to the switching frequency, having a model based only on the fundamental frequency. In the case of the Buck-boost converter, the fundamental frequency is 0 Hz since it is a DC/DC converter.

By applying (2.12) to the precedent equation system (2.6), we obtain:

$$\begin{cases} \frac{d\langle V_s \rangle}{dt} = \frac{\langle s(t) \rangle - 1}{C} \langle i_L \rangle - \frac{1}{RC} \langle V_s \rangle \\ \frac{d\langle i_L \rangle}{dt} = \frac{1}{L} \langle s(t) \rangle \cdot V_{in} + \frac{1 - \langle s(t) \rangle}{L} \langle V_s \rangle \end{cases} \quad (2.13)$$

To obtain the switching function average value, the PWM strategy must be considered to define the $s(t)$ function to be integrated.

$$\langle s(t) \rangle = \frac{1}{T_{sw}} \int_0^{T_{sw}} s(\tau) d\tau \quad (2.14)$$

The PWM strategy compares a sawtooth waveform with f_{sw} frequency (carrier) to the duty cycle α value (modulation signal) every sampling period as in the example of Figure 2.2.

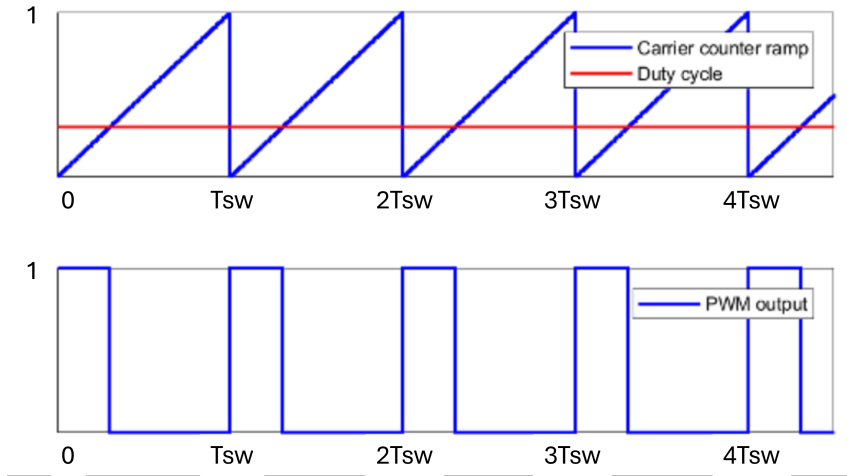


Figure 2.2: PWM logic.

The carrier with sawtooth waveform expression for $0 < t < T_{sw}$ is:

$$tr(t) = \frac{1}{T_{sw}} t \quad (2.15)$$

With this expression, we can find the time where $tr(t)$ becomes higher than α :

$$tr(t_\alpha) = \alpha = \frac{1}{T_{sw}} t_\alpha \rightarrow t_\alpha = \alpha T_{sw} \quad (2.16)$$

Knowing that the PWM output is either 1 when the duty cycle is greater than the carrier or 0 otherwise, we can derive that the $s(t)$ expression within one period ($0 < t < T_{sw}$) is:

$$s(t) = \mathbf{1}_{\{t < t_\alpha\}} \quad (2.17)$$

where $\mathbf{1}_{\{t < t_\alpha\}}$ is a step function, meaning that:

$$s(t) = \begin{cases} 1, & \text{if } t < t_\alpha = \alpha T_{sw}, \\ 0, & \text{otherwise.} \end{cases} \quad (2.18)$$

Therefore, by applying (2.18) definition on (2.14), the switching function average value over the switching period T_{sw} is equal to the duty cycle α :

$$\langle s(t) \rangle = \frac{1}{T_{sw}} \int_0^{T_{sw}} s(\tau) d\tau = \frac{1}{T_{sw}} \int_0^{\alpha T_{sw}} 1 d\tau + \frac{1}{T_{sw}} \int_{\alpha T_{sw}}^{T_{sw}} 0 d\tau = \frac{1}{T_{sw}} \cdot \alpha T_{sw} = \alpha \quad (2.19)$$

Applying (2.19), the system of (2.20) becomes:

$$\begin{cases} \frac{d\langle V_s \rangle}{dt} = \frac{\alpha-1}{C} \langle i_L \rangle - \frac{1}{RC} \langle V_s \rangle \\ \frac{d\langle i_L \rangle}{dt} = \frac{1}{L} \alpha \cdot V_{in} + \frac{1-\alpha}{L} \langle V_s \rangle \end{cases} \quad (2.20)$$

Then, the state-space representation can be written as follows:

$$\frac{d}{dt} x_{ave} = A_{ave} x_{ave} + B_{ave} u_{ave} \quad (2.21)$$

with,

$$x_{ave} = \begin{bmatrix} \langle i_L \rangle \\ \langle V_s \rangle \end{bmatrix} \quad (2.22)$$

$$u_{ave} = \alpha \langle V_{in} \rangle \quad (2.23)$$

$$A_{ave} = \begin{bmatrix} 0 & \frac{1-\alpha}{L} \\ \frac{\alpha-1}{C} & -\frac{1}{RC} \end{bmatrix} \quad (2.24)$$

$$B_{ave} = \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} \quad (2.25)$$

2.3 Harmonic State-Space detailed model (HSS)

The HSS model equation equivalent to the state dynamic equation on the TSS model is:

$$sX = (\mathcal{A} - N)X + \mathcal{B}U \quad (2.26)$$

with the following equivalent state and input vectors, and state and control matrices,

$$X = [x_{-N} \quad \dots \quad x_{-1} \quad x_0 \quad x_1 \quad \dots \quad x_N]^T \quad (2.27)$$

$$U = [u_{-N} \quad \dots \quad u_{-1} \quad u_0 \quad u_1 \quad \dots \quad u_N]^T \quad (2.28)$$

$$\mathcal{A} = \text{toeplitz}(A_{-2N}, \dots, A_{-1}, A_0, A_1, \dots, A_{2N}) \quad (2.29)$$

$$\mathcal{B} = \text{toeplitz}(B_{-2N}, \dots, B_{-1}, B_0, B_1, \dots, B_{2N}) \quad (2.30)$$

From the detailed Time State-Space (TSS) representation developed before, we can derive the Harmonic State-Space (HSS) detailed model of the Buck-Boost converter by writing all the matrices and vectors in terms of Fourier series. For this, we use the formula of the decomposition of a signal in terms of Fourier series:

$$x(t) = \sum_{k=-N}^N x_k e^{jk\omega_0 t} \quad (2.31)$$

Then, to define the HSS model, it is sufficient to compute the coefficients x_k of the Fourier series using:

$$x_k = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-jk\omega_0 t} dt \quad (2.32)$$

for all the matrices and vectors of the TSS model.

2.3.1 Matrix X

The definition of X is:

$$X = [x_{-N} \quad \dots \quad x_{-1} \quad x_0 \quad x_1 \quad \dots \quad x_N]^T \quad (2.33)$$

with,

$$X_k = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-jk\omega_0 t} dt \quad (2.34)$$

Knowing the state vector is $x(t) = \begin{bmatrix} i_L(t) \\ V_s(t) \end{bmatrix}$,

$$X_k = \frac{1}{T_0} \int_0^{T_0} \begin{bmatrix} i_L(t) e^{-jk\omega_0 t} \\ V_s(t) e^{-jk\omega_0 t} \end{bmatrix} dt \quad (2.35)$$

Therefore, we have that:

$$X_k = \begin{bmatrix} I_{L,k} \\ V_{s,k} \end{bmatrix} \quad (2.36)$$

The determination of $I_{L,k}$ and $V_{s,k}$ is done directly using the fft function in MATLAB. Then, the state vector X for the HSS model can be defined:

$$X = [I_{L,-N} \quad V_{s,-N} \quad \dots \quad I_{L,0} \quad V_{s,0} \quad \dots \quad I_{L,N} \quad V_{s,N}]^T \quad (2.37)$$

2.3.2 Matrix U

The definition of U is:

$$U = [u_{-N} \quad \dots \quad u_{-1} \quad u_0 \quad u_1 \quad \dots \quad u_N]^T \quad (2.38)$$

with,

$$U_k = \frac{1}{T_0} \int_0^{T_0} u(t) e^{-jk\omega_0 t} dt \quad (2.39)$$

Knowing the input u is $u(t) = s(t)V_{in}$,

$$U_k = \frac{1}{T_0} \int_0^{T_0} s(t) V_{in} e^{-jk\omega_0 t} dt \quad (2.40)$$

Therefore, we have that:

$$U_k = V_{in} \cdot S_k \quad (2.41)$$

The determination of S_k is done directly using the fft function in MATLAB. Then, the input vector U for the HSS model can be defined:

$$U = V_{in} \cdot [S_{-N} \quad \dots \quad S_0 \quad \dots \quad S_N]^T \quad (2.42)$$

2.3.3 Matrix $\mathcal{A}$

The definition of $\mathcal{A}$ is:

$$\mathcal{A} = \text{toeplitz}(A_{-2N}, \dots, A_{-1}, A_0, A_1, \dots, A_{2N}) \quad (2.43)$$

with,

$$A_k = \frac{1}{T_0} \int_0^{T_0} A(t) e^{-jk\omega_0 t} dt \quad (2.44)$$

Knowing the state matrix is $A(t) = \begin{bmatrix} 0 & \frac{1-s(t)}{L} \\ \frac{s(t)-1}{C} & -\frac{1}{RC} \end{bmatrix}$,

$$A_k = \frac{1}{T_0} \int_0^{T_0} \begin{bmatrix} 0 & \frac{1-s(t)}{L} \\ \frac{s(t)-1}{C} & -\frac{1}{RC} \end{bmatrix} e^{-jk\omega_0 t} dt \quad (2.45)$$

To define A_k , we use the following property:

$$\int_0^{T_0} e^{jm\omega_0 t} e^{-jn\omega_0 t} dt = \delta_{mn} T_0 \quad (2.46)$$

where the Kronecker Delta is the impulse function defined as:

$$\delta_{mn} = \begin{cases} 1, & \text{if } m = n, \\ 0, & \text{otherwise.} \end{cases} \quad (2.47)$$

Manipulating (2.45),

$$A_k = \frac{1}{T_0} \int_0^{T_0} \begin{bmatrix} 0 & \frac{1-s(t)}{L} e^{-jk\omega_0 t} \\ \frac{s(t)-1}{C} e^{-jk\omega_0 t} & -\frac{1}{RC} \cdot e^{-jk\omega_0 t} \end{bmatrix} dt \quad (2.48)$$

Integrating each term of the matrix individually we obtain A_k . For the (1,2) term:

$$\frac{1}{T_0} \int_0^{T_0} \frac{s(t)-1}{C} e^{-jk\omega_0 t} dt = \frac{1}{T_0} \left(\int_0^{T_0} \frac{s(t)}{C} e^{-jk\omega_0 t} dt - \int_0^{T_0} \frac{1}{C} e^{-jk\omega_0 t} dt \right) \quad (2.49)$$

$$= \frac{1}{T_0} \left(\int_0^{T_0} \frac{s(t)}{C} e^{-jk\omega_0 t} dt - \int_0^{T_0} \frac{e^{j0\omega_0 t}}{C} e^{-jk\omega_0 t} dt \right) \quad (2.50)$$

Applying property (2.46) and the Fourier series coefficients definition (2.32),

$$= \frac{S_k}{C} - \frac{\delta_{0k} \cdot T_0}{C \cdot T_0} = \frac{S_k - \delta_{0k}}{C} \quad (2.51)$$

For the (2,1) term:

$$\frac{1}{T_0} \int_0^{T_0} \frac{1-s(t)}{L} e^{-jk\omega_0 t} dt = \frac{1}{T_0} \left(\int_0^{T_0} \frac{1}{L} e^{-jk\omega_0 t} dt - \int_0^{T_0} \frac{s(t)}{L} e^{-jk\omega_0 t} dt \right) \quad (2.52)$$

$$= \frac{1}{T_0} \left(\int_0^{T_0} \frac{e^{j0\omega_0 t}}{L} e^{-jk\omega_0 t} - \int_0^{T_0} \frac{s(t)}{L} e^{-jk\omega_0 t} \right) dt \quad (2.53)$$

Applying property (2.46) and the Fourier series coefficients definition (2.32),

$$= \frac{\delta_{0k} \cdot T_0}{L \cdot T_0} - \frac{S_k}{L} = \frac{\delta_{0k} - S_k}{L} \quad (2.54)$$

For the (2,2) term:

$$\frac{1}{T_0} \int_0^{T_0} -\frac{1}{RC} \cdot e^{-jk\omega_0 t} dt = \frac{1}{T_0} \int_0^{T_0} -\frac{1}{RC} \cdot 1 \cdot e^{-jk\omega_0 t} dt = \frac{1}{T_0} \int_0^{T_0} -\frac{1}{RC} \cdot e^{j0\omega_0 t} \cdot e^{-jk\omega_0 t} dt \quad (2.55)$$

Applying property (2.46) and the Fourier series coefficients definition (2.32),

$$= -\frac{1}{RC} \frac{\delta_{0k} \cdot T_0}{T_0} = -\frac{1}{RC} \delta_{0k} \quad (2.56)$$

Finally, A_k can be written:

$$A_k = \begin{bmatrix} 0 & \frac{\delta_{0k} - S_k}{L} \\ \frac{S_k - \delta_{0k}}{C} & -\frac{1}{RC} \delta_{0k} \end{bmatrix} \quad (2.57)$$

Meaning that:

$$A_k = \begin{cases} \begin{bmatrix} 0 & \frac{1-S_0}{L} \\ \frac{S_0-1}{C} & -\frac{1}{RC} \end{bmatrix}, & \text{if } k = 0, \\ \begin{bmatrix} 0 & \frac{S_k}{L} \\ \frac{S_k}{C} & 0 \end{bmatrix}, & \text{otherwise.} \end{cases} \quad (2.58)$$

The determination of S_k is done directly using the fft function in MATLAB as used to compute both A_k and U_k .

The final matrix $\mathcal{A}$ is going to have the form:

$$\mathcal{A} = \begin{bmatrix} A_0 & A_{-1} & \cdots & \cdots & \cdots & \cdots & A_{-2N} \\ A_1 & \ddots & \ddots & & & & \vdots \\ \vdots & \ddots & \ddots & \ddots & & & \vdots \\ \vdots & & \ddots & \ddots & \ddots & & \vdots \\ \vdots & & & \ddots & \ddots & \ddots & \vdots \\ \vdots & & & & \ddots & \ddots & A_{-1} \\ A_{2N} & \cdots & \cdots & \cdots & \cdots & A_1 & A_0 \end{bmatrix} \quad (2.59)$$

For example, we can compile $\mathcal{A}$ for $N = 1$:

$$\mathcal{A} = \begin{bmatrix} \begin{bmatrix} 0 & \frac{1-S_0}{L} \\ \frac{S_0-1}{C} & -\frac{1}{RC} \end{bmatrix} & \begin{bmatrix} 0 & \frac{S_{-1}}{L} \\ \frac{S_{-1}}{C} & 0 \end{bmatrix} & \begin{bmatrix} 0 & \frac{S_{-2}}{L} \\ \frac{S_{-2}}{C} & 0 \end{bmatrix} \\ \begin{bmatrix} 0 & \frac{S_1}{L} \\ \frac{S_1}{C} & 0 \end{bmatrix} & \begin{bmatrix} 0 & \frac{1-S_0}{L} \\ \frac{S_0-1}{C} & -\frac{1}{RC} \end{bmatrix} & \begin{bmatrix} 0 & \frac{S_{-1}}{L} \\ \frac{S_{-1}}{C} & 0 \end{bmatrix} \\ \begin{bmatrix} 0 & \frac{S_2}{L} \\ \frac{S_2}{C} & 0 \end{bmatrix} & \begin{bmatrix} 0 & \frac{S_1}{L} \\ \frac{S_1}{C} & 0 \end{bmatrix} & \begin{bmatrix} 0 & \frac{1-S_0}{L} \\ \frac{S_0-1}{C} & -\frac{1}{RC} \end{bmatrix} \end{bmatrix} \quad (2.60)$$

$$\mathcal{A} = \begin{bmatrix} 0 & \frac{1-S_0}{L} & 0 & \frac{S_{-1}}{L} & 0 & \frac{S_{-2}}{L} \\ \frac{S_0-1}{C} & -\frac{1}{RC} & \frac{S_{-1}}{C} & 0 & \frac{S_{-2}}{C} & 0 \\ 0 & \frac{1-S_1}{L} & 0 & \frac{1-S_0}{L} & 0 & \frac{1-S_{-1}}{L} \\ \frac{S_1}{C} & 0 & \frac{S_0-1}{C} & -\frac{1}{RC} & \frac{S_{-1}}{C} & 0 \\ 0 & \frac{S_2}{L} & 0 & \frac{S_1}{L} & 0 & \frac{1-S_0}{L} \\ \frac{S_2}{C} & 0 & \frac{S_1}{C} & 0 & \frac{S_0-1}{C} & -\frac{1}{RC} \end{bmatrix} \quad (2.61)$$

2.3.4 Matrix $\mathcal{B}$

The definition of $\mathcal{B}$ is:

$$\mathcal{B} = \text{toeplitz}(B_{-2N}, \dots, B_{-1}, B_0, B_1, \dots, B_{2N}) \quad (2.62)$$

with,

$$B_k = \frac{1}{T_0} \int_0^{T_0} B(t) e^{-jk\omega_0 t} dt \quad (2.63)$$

Knowing the input matrix is $B = \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix}$ and not dependent of time,

$$B_k = \frac{1}{T_0} \int_0^{T_0} \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} e^{-jk\omega_0 t} dt \quad (2.64)$$

Manipulating (2.64),

$$B_k = \frac{1}{T_0} \int_0^{T_0} \begin{bmatrix} \frac{1}{L} 1 \cdot e^{-jk\omega_0 t} \\ 0 \end{bmatrix} dt = \frac{1}{T_0} \int_0^{T_0} \begin{bmatrix} \frac{1}{L} e^{j0\omega_0 t} e^{-jk\omega_0 t} \\ 0 \end{bmatrix} dt \quad (2.65)$$

Finally, applying the property (2.46) in (2.66):

$$B_k = \begin{bmatrix} \frac{1}{L} \delta_{0k} \\ 0 \end{bmatrix} \quad (2.66)$$

Meaning that:

$$B_k = \begin{cases} \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix}, & \text{if } k = 0, \\ 0_{2 \times 1}, & \text{otherwise.} \end{cases} \quad (2.67)$$

The final matrix $\mathcal{B}$ is going to have the form:

$$\mathcal{B} = \begin{bmatrix} B_0 & 0_{2 \times 1} & \cdots & \cdots & \cdots & \cdots & 0_{2 \times 1} \\ 0_{2 \times 1} & \ddots & \ddots & & & & \vdots \\ \vdots & \ddots & \ddots & \ddots & & & \vdots \\ \vdots & & \ddots & \ddots & \ddots & & \vdots \\ \vdots & & & \ddots & \ddots & \ddots & \vdots \\ \vdots & & & & \ddots & \ddots & 0_{2 \times 1} \\ 0_{2 \times 1} & \cdots & \cdots & \cdots & \cdots & 0_{2 \times 1} & B_0 \end{bmatrix} \quad (2.68)$$

For example, we can compile $\mathcal{B}$ for $N = 1$:

$$\mathcal{B} = \begin{bmatrix} \begin{bmatrix} \frac{1}{L} \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L} \\ 0 \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{L} \\ 0 \end{bmatrix} \end{bmatrix} \quad (2.69)$$

$$\mathcal{B} = \begin{bmatrix} \frac{1}{L} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \frac{1}{L} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{1}{L} \\ 0 & 0 & 0 \end{bmatrix} \quad (2.70)$$

2.4 Harmonic State-Space average model

The HSS average model is derived similarly using the equations of the average TSS model.

2.4.1 Matrix X_{ave}

To be able to define X_{ave} , we have to first define:

$$X_{ave,k} = \frac{1}{T_0} \int_0^{T_0} x_{ave} e^{-jk\omega_0 t} dt \quad (2.71)$$

Knowing the state vector is $x_{ave} = \begin{bmatrix} \langle i_L \rangle \\ \langle V_s \rangle \end{bmatrix}$,

$$X_{ave,k} = \frac{1}{T_0} \int_0^{T_0} \begin{bmatrix} \langle i_L \rangle e^{-jk\omega_0 t} \\ \langle V_s \rangle e^{-jk\omega_0 t} \end{bmatrix} dt \quad (2.72)$$

Knowing that $\langle i_L \rangle$ and $\langle V_s \rangle$ are constant values, and manipulating (2.72),

$$X_{ave,k} = \frac{1}{T_0} \int_0^{T_0} \begin{bmatrix} \langle i_L \rangle \cdot 1 \cdot e^{-jk\omega_0 t} \\ \langle V_s \rangle \cdot 1 \cdot e^{-jk\omega_0 t} \end{bmatrix} dt = \frac{1}{T_0} \int_0^{T_0} \begin{bmatrix} \langle i_L \rangle \cdot 1 \cdot e^{j0\omega_0 t} e^{-jk\omega_0 t} \\ \langle V_s \rangle \cdot 1 \cdot e^{j0\omega_0 t} e^{-jk\omega_0 t} \end{bmatrix} dt \quad (2.73)$$

Finally, applying the property (2.46) in (2.73):

$$X_{ave,k} = \begin{bmatrix} \langle i_L \rangle \\ \langle V_s \rangle \end{bmatrix} \delta_{0k} \quad (2.74)$$

Meaning that:

$$X_{ave,k} = \begin{cases} \begin{bmatrix} \langle i_L \rangle \\ \langle V_s \rangle \end{bmatrix}, & \text{if } k = 0, \\ 0_{2 \times 1}, & \text{otherwise.} \end{cases} \quad (2.75)$$

Then, the state vector X_{ave} for the average HSS model can be defined:

$$X_{ave} = [0 \quad \dots \quad 0 \quad \langle i_L \rangle \quad \langle V_s \rangle \quad 0 \quad \dots \quad 0]^T \quad (2.76)$$

2.4.2 Matrix U_{ave}

To be able to define U_{ave} , we have to first define:

$$U_{ave,k} = \frac{1}{T_0} \int_0^{T_0} u_{ave} e^{-jk\omega_0 t} dt \quad (2.77)$$

Knowing the input is $u_{ave} = \alpha V_{in}$,

$$U_{ave,k} = \frac{1}{T_0} \int_0^{T_0} \alpha < V_{in} > e^{-jk\omega_0 t} dt \quad (2.78)$$

Manipulating (2.78),

$$U_{ave,k} = \frac{1}{T_0} \int_0^{T_0} \alpha < V_{in} > 1 \cdot e^{-jk\omega_0 t} dt = \frac{1}{T_0} \int_0^{T_0} \alpha < V_{in} > e^{j0\omega_0 t} e^{-jk\omega_0 t} dt \quad (2.79)$$

Finally, applying the property (2.46) in (2.80):

$$U_{ave,k} = \alpha < V_{in} > \delta_{0k} \quad (2.80)$$

Meaning that:

$$U_{ave,k} = \begin{cases} \alpha < V_{in} >, & \text{if } k = 0, \\ 0, & \text{otherwise.} \end{cases} \quad (2.81)$$

Then, the input vector U_{ave} for the average HSS model can be defined:

$$U_{ave} = [0 \quad \dots \quad 0 \quad \alpha < V_{in} > \quad 0 \quad \dots \quad 0]^T \quad (2.82)$$

2.4.3 Matrix $\mathcal{A}_{ave}$

To be able to define $\mathcal{A}_{ave}$, we have to first define:

$$A_{ave,k} = \frac{1}{T_0} \int_0^{T_0} A_{ave} e^{-jk\omega_0 t} dt \quad (2.83)$$

Knowing the state matrix is $A_{ave} = \begin{bmatrix} 0 & \frac{1-\alpha}{L_1} \\ \frac{\alpha-1}{C} & -\frac{1}{RC} \end{bmatrix}$,

$$A_{ave,k} = \frac{1}{T_0} \int_0^{T_0} \begin{bmatrix} 0 & \frac{1-\alpha}{L_1} \\ \frac{\alpha-1}{C} & -\frac{1}{RC} \end{bmatrix} e^{-jk\omega_0 t} dt \quad (2.84)$$

Manipulating (2.84),

$$A_{ave,k} = \frac{1}{T_0} \int_0^{T_0} \begin{bmatrix} 0 & \frac{1-\alpha}{L_1} \\ \frac{\alpha-1}{C} & -\frac{1}{RC} \end{bmatrix} 1 \cdot e^{-jk\omega_0 t} dt \quad (2.85)$$

$$A_{ave,k} = \frac{1}{T_0} \int_0^{T_0} \begin{bmatrix} 0 & \frac{1-\alpha}{L_1} \\ \frac{\alpha-1}{C} & -\frac{1}{RC} \end{bmatrix} e^{j0\omega_0 t} e^{-jk\omega_0 t} dt \quad (2.86)$$

Finally, applying the property (2.46):

$$A_{ave,k} = \begin{bmatrix} 0 & \frac{1-\alpha}{L_1} \\ \frac{\alpha-1}{C} & -\frac{1}{RC} \end{bmatrix} \delta_{0k} \quad (2.87)$$

Meaning that:

$$A_{ave,k} = \begin{cases} \begin{bmatrix} 0 & \frac{1-\alpha}{L} \\ \frac{\alpha-1}{C} & -\frac{1}{RC} \end{bmatrix}, & \text{if } k = 0, \\ 0_2, & \text{otherwise.} \end{cases} \quad (2.88)$$

The final matrix $\mathcal{A}_{ave}$ is going to have the form:

$$\mathcal{A}_{ave} = \begin{bmatrix} A_{ave,0} & 0_2 & \cdots & \cdots & \cdots & \cdots & 0_2 \\ A_1 & \ddots & \ddots & & & & \vdots \\ \vdots & \ddots & \ddots & \ddots & & & \vdots \\ \vdots & & \ddots & \ddots & \ddots & & \vdots \\ \vdots & & & \ddots & \ddots & \ddots & \vdots \\ \vdots & & & & \ddots & \ddots & 0_2 \\ 0_2 & \cdots & \cdots & \cdots & \cdots & 0_2 & A_{ave,0} \end{bmatrix} \quad (2.89)$$

For example, we can compile $\mathcal{A}_{ave}$ for $N = 1$:

$$\mathcal{A}_{ave} = \begin{bmatrix} \begin{bmatrix} 0 & \frac{1-\alpha}{L} \\ \frac{\alpha-1}{C} & -\frac{1}{RC} \end{bmatrix} & \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} & \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \\ \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} & \begin{bmatrix} 0 & \frac{1-\alpha}{L} \\ \frac{\alpha-1}{C} & -\frac{1}{RC} \end{bmatrix} & \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \\ \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} & \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} & \begin{bmatrix} 0 & \frac{1-\alpha}{L} \\ \frac{\alpha-1}{C} & -\frac{1}{RC} \end{bmatrix} \end{bmatrix} \quad (2.90)$$

$$\mathcal{A}_{ave} = \begin{bmatrix} 0 & \frac{1-\alpha}{L} & 0 & 0 & 0 & 0 \\ \frac{\alpha-1}{C} & -\frac{1}{RC} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1-\alpha}{L} & 0 & 0 \\ 0 & 0 & \frac{\alpha-1}{C} & -\frac{1}{RC} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1-\alpha}{L} \\ 0 & 0 & 0 & 0 & \frac{\alpha-1}{C} & -\frac{1}{RC} \end{bmatrix} \quad (2.91)$$

2.4.4 Matrix $\mathcal{B}_{ave}$

To be able to define $\mathcal{B}_{ave}$, we have to first define:

$$B_{ave,k} = \frac{1}{T_0} \int_0^{T_0} B_{ave}(t) e^{-jk\omega_0 t} dt \quad (2.92)$$

Knowing the input matrix is $B_{ave} = \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix}$ and is the same as the development for the detailed model, we can deduce directly that:

$$B_{ave,k} = \begin{bmatrix} \frac{1}{L} \delta_{0k} \\ 0 \end{bmatrix} \quad (2.93)$$

Meaning that:

$$B_{ave,k} = \begin{cases} \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix}, & \text{if } k = 0, \\ 0_{2 \times 1}, & \text{otherwise.} \end{cases} \quad (2.94)$$

The final matrix $\mathcal{B}_{ave}$ is going to have the form:

$$\mathcal{B}_{ave} = \begin{bmatrix} B_{ave,0} & 0_{2 \times 1} & \cdots & \cdots & \cdots & \cdots & 0_{2 \times 1} \\ 0_{2 \times 1} & \ddots & \ddots & & & & \vdots \\ \vdots & & \ddots & \ddots & \ddots & & \vdots \\ \vdots & & & \ddots & \ddots & \ddots & \vdots \\ \vdots & & & & \ddots & \ddots & \vdots \\ \vdots & & & & & \ddots & \vdots \\ \vdots & & & & & & 0_{2 \times 1} \\ 0_{2 \times 1} & \cdots & \cdots & \cdots & \cdots & 0_{2 \times 1} & B_{ave,0} \end{bmatrix} \quad (2.95)$$

For example, we can compile $\mathcal{B}_{ave}$ for $N = 1$:

$$\mathcal{B}_{ave} = \begin{bmatrix} \frac{1}{L} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \frac{1}{L} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{1}{L} \\ 0 & 0 & 0 \end{bmatrix} \quad (2.96)$$

Chapter 3

Simulation and results

3.1 Detailed models comparison

To validate both the TSS and HSS detailed (switching) models, the Simulink simulation of Figure 3.1 is used. In this simulation, the TSS and the HSS models simulation are compared to the circuit simulation results.

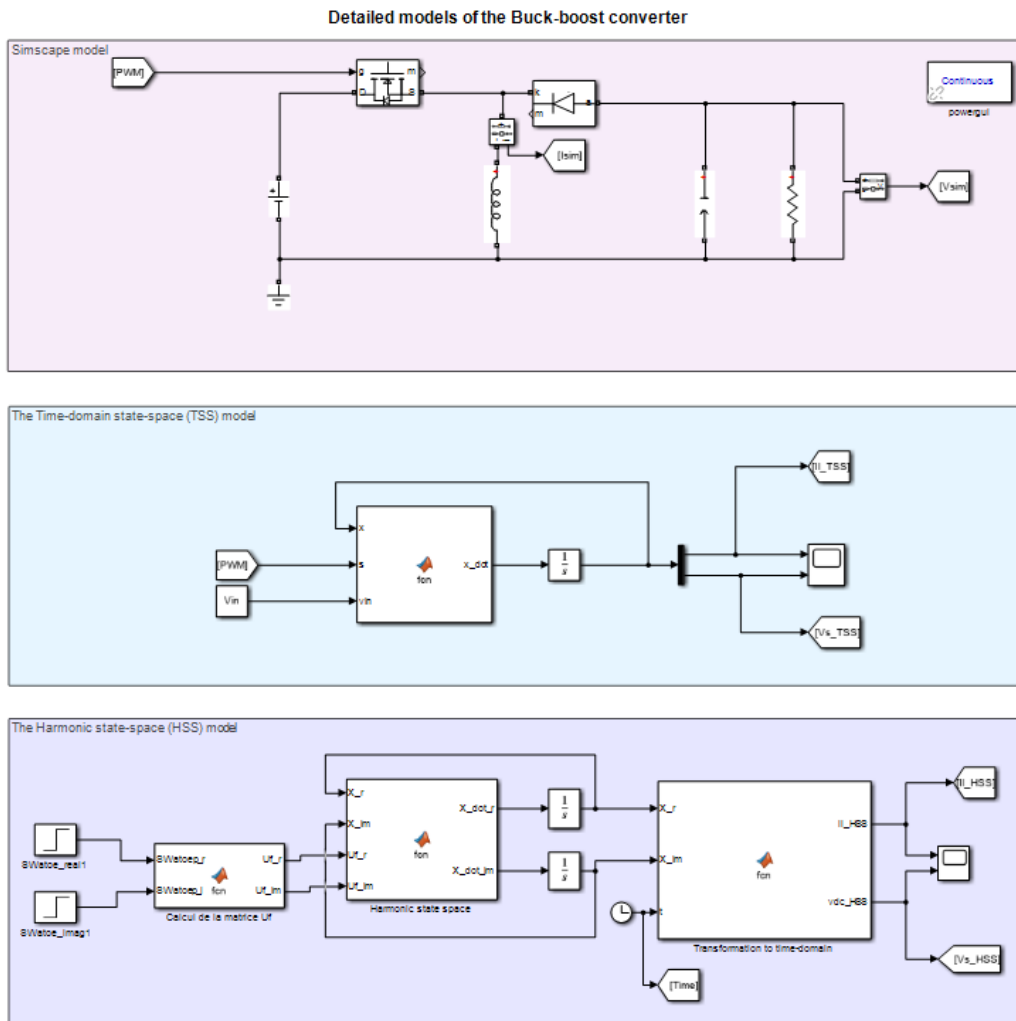


Figure 3.1: Simulink simulation of detailed models.

The parameters that were used in this comparison are those of [Table 3.1](#) and were chosen as follows.

Table 3.1: Simulation parameters

Symbol	Parameter	Value
f_{sw}	Switching frequency	100 Hz
V_{in}	Voltage input	12 V
$\langle V_s \rangle$	Mean voltage output	-5 V
P_{in}	Power input	7 W
ΔV_s	Output voltage peak-to-peak amplitude	1,16 V
ΔI_L	Inductor current peak-to-peak amplitude	2,08 A
α	Duty cycle	$\frac{5}{17}$
R	Resistance	3,57 Ω
L	Inductance	16,968 mH
C	Capacitance	3,55 mF

3.1.1 Parameters choice

The circuit resistance, inductance and capacitance of the Buck-boost converter are not given, but instead they are chosen from the other specifications.

The resistance is defined according to the desired converter output voltage and power:

$$R = \frac{\langle V_s \rangle^2}{P_s} \approx \frac{\langle V_s \rangle^2}{P_{in}} = 3,57 \Omega \quad (3.1)$$

The inductance is defined using the peak-to-peak inductor current formula from [\(1.12\)](#):

$$L = -\frac{\langle V_s \rangle (1 - \alpha)}{f_{sw} \Delta I} = 16,968 \text{ mH} \quad (3.2)$$

The capacitor capacitance is defined using that, by the KCL, the capacitor current is:

$$i_C = -i_{k2} - i_R \quad (3.3)$$

Applying [\(3.3\)](#) to the capacitor dynamic equation [\(1.6\)](#):

$$C \frac{d}{dt} V_s = -(1 - s(t)) \cdot i_L - i_R \quad (3.4)$$

Integrating over 0 to αT_{sw} :

$$\int_0^{\alpha T_{sw}} C \frac{d}{dt} V_s dt = \int_0^{\alpha T_{sw}} -(1 - s(t)) \cdot i_L - i_R dt \quad (3.5)$$

Using that, for t in $0 < t < \alpha T_{sw}$, $s(t) = 1$:

$$C(V_{s,min} - V_{s,max}) = \int_0^{\alpha T_{sw}} -i_R dt = \int_0^{\alpha T_{sw}} -\frac{V_s}{R} dt = -\frac{V_s}{R} \alpha T_{sw} \quad (3.6)$$

Therefore,

$$\Delta V_s = -\frac{V_s \alpha T_{sw}}{RC} \quad (3.7)$$

We are able to design the capacitor by isolating the capacitance C in equation (3.8), that leads us to:

$$C = -\frac{\langle V_s \rangle \alpha T_{sw}}{R \Delta V_s} = 3,55 \text{ mF} \quad (3.8)$$

3.1.2 Simulation results

Figure 3.3 shows the comparison of the Simulink simulation results in steady-state for the three detailed models: circuit model, TSS model and HSS model. The HSS model in this comparison used the number of harmonics $N = 30$.

We observe that the three models give practically the same result, meaning that the TSS and the HSS models are a good representation of the converter operation.

Also, the measured Δi_L and ΔV_s amplitudes are equal to 2,078 A and 1,24 V respectively, which is consistent to the specifications defined in Table 3.1.

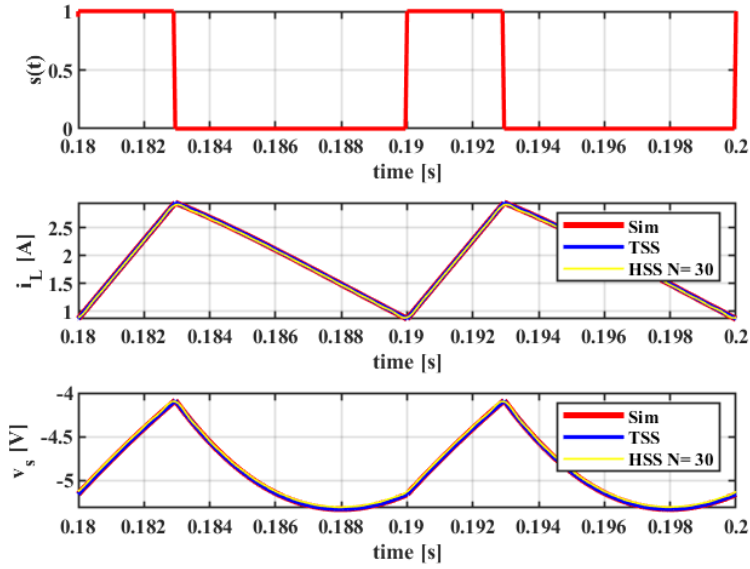


Figure 3.2: Simulink simulation steady-state results for the detailed models.

Figure 3.3 shows the comparison of the Simulink simulation transient results for the three detailed models. From these results, we can conclude that the TSS and HSS models are also representative of the converter operation in transient regime.

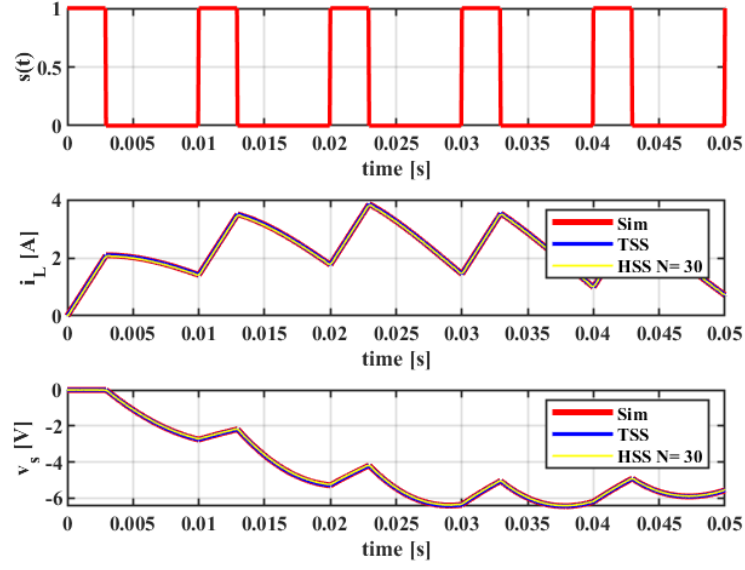


Figure 3.3: Simulink simulation transient results for the detailed models.

3.2 Average models comparison

To validate both the TSS and HSS average models, the Simulink simulation of [Figure 3.4](#) is used. In this simulation, the TSS and the HSS models simulation results are compared.

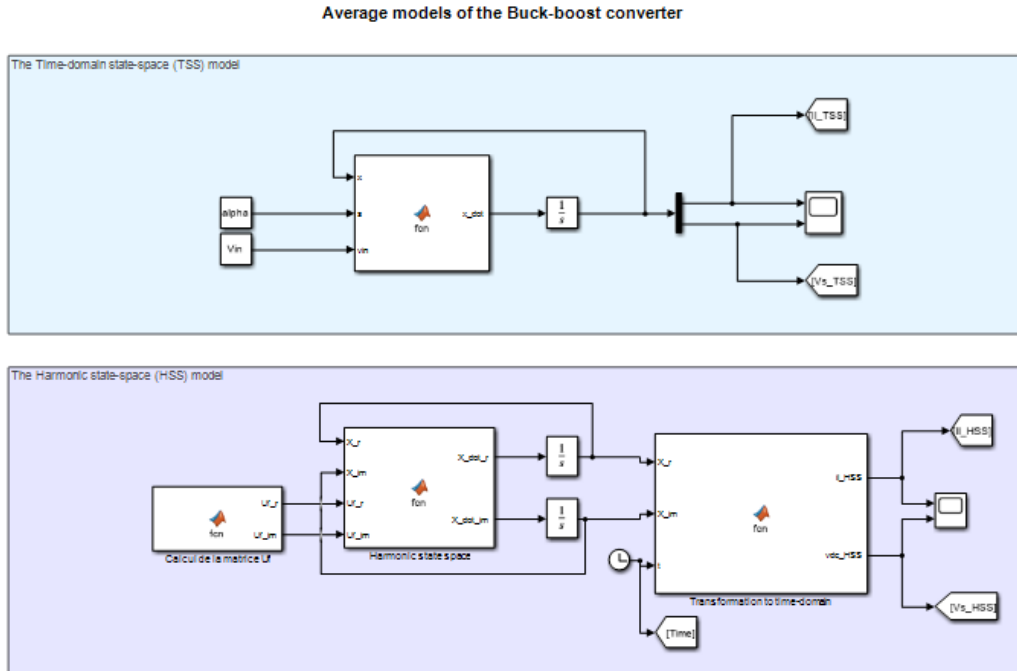


Figure 3.4: Simulink simulation of average models.

The parameters that were used in this comparison are those of [Table 3.1](#) and are the

same for the detailed models.

3.2.1 Simulation results

Figure 3.5 shows the comparison of the Simulink simulation results for the TSS and HSS average models. The HSS average model in this comparison used the number of harmonics $N = 30$ as in the detailed model.

We observe that the two models give practically the same result, which is coherent since the HSS average model is derived from the TSS average model. The state variables i_L and V_s have no longer variation within one switching period such that the switching frequency is "taken out" of the results, as previously explained in section 2.2.

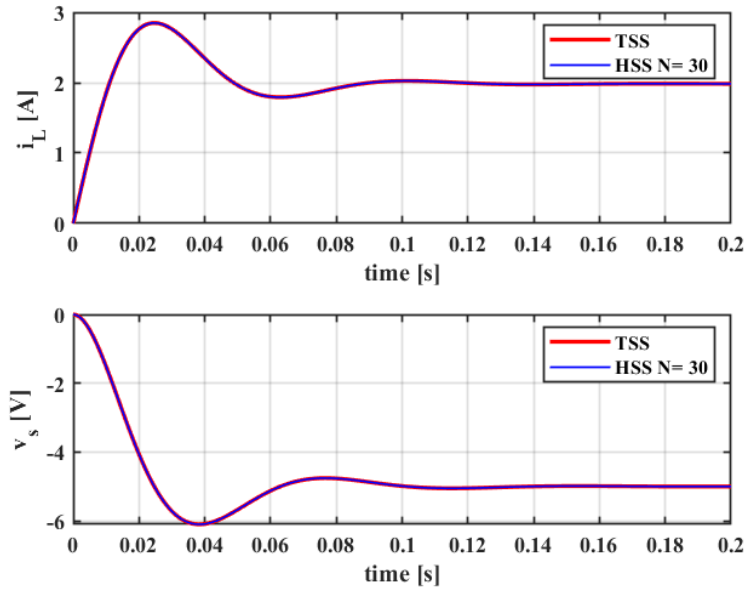


Figure 3.5: Simulink simulation results for the average models.

This simplification is clearer observed in Figure 3.6 that shows the comparison of the Simulink simulation results for the TSS and HSS average models with the detailed circuit Simulink model. The comparison also shows that the average models follows the detailed model signals mean value over one T_{sw} period.

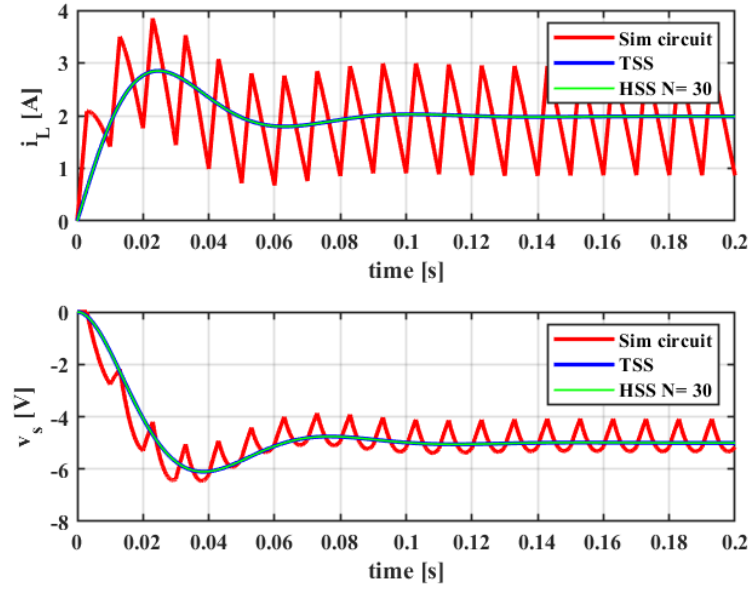


Figure 3.6: Average model simplification result.

Bibliography

- [1] Y. Nadjai, L. Quéval, "Modèle HSS du convertisseur Buck", Rapport projet initiation à la recherche, Paris-Saclay, France.